

Active Directory Lab – Windows Server 2022

Design, Implementation and Validation

A hands-on Active Directory infrastructure lab covering domain services, security hardening and high availability.

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September 2026

AD-LAB-V1

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1. Introduction

This document provides a detailed overview of the design, implementation, and validation of a personal Active Directory lab built on Windows Server 2022 and Oracle VirtualBox. The goal of this lab is to simulate a small enterprise environment in a controlled setting, demonstrating core identity, networking, and security concepts.

The lab domain, LAB.local, includes multiple domain controllers (DC01, DC02, RODC01), DNS and DHCP services, Group Policy-based hardening, and both Windows and Linux domain-joined clients. The configuration focuses on realistic, production-like scenarios such as redundancy, DHCP failover, and the use of a Read-Only Domain Controller (RODC) for branch-office scenarios.

This report is intended as technical documentation that can be used by recruiters, hiring managers, or peers to review the lab design and validate the configuration steps. It complements the high-level README in the GitHub repository by providing screenshots, command outputs, and detailed explanations for each component.

2. Lab Overview

2.1. Virtualization Platform and Network

The lab is hosted on Oracle VirtualBox, running on a single physical host. All virtual machines are connected to an isolated internal network for domain traffic, with an additional NAT-adaptor used to provide controlled internet access where required.

The internal network is configured as 192.168.56.0/24, with static IP addresses assigned to the domain controllers and DHCP used for client devices. DC01 and DC02 are configured with two network interfaces: one connected to the internal domain network and one connected through NAT. RODC01 uses the internal network for domain communication and does not require direct external connectivity. This design separates internal Active Directory traffic from external access while maintaining connectivity for administrative and update-related tasks.

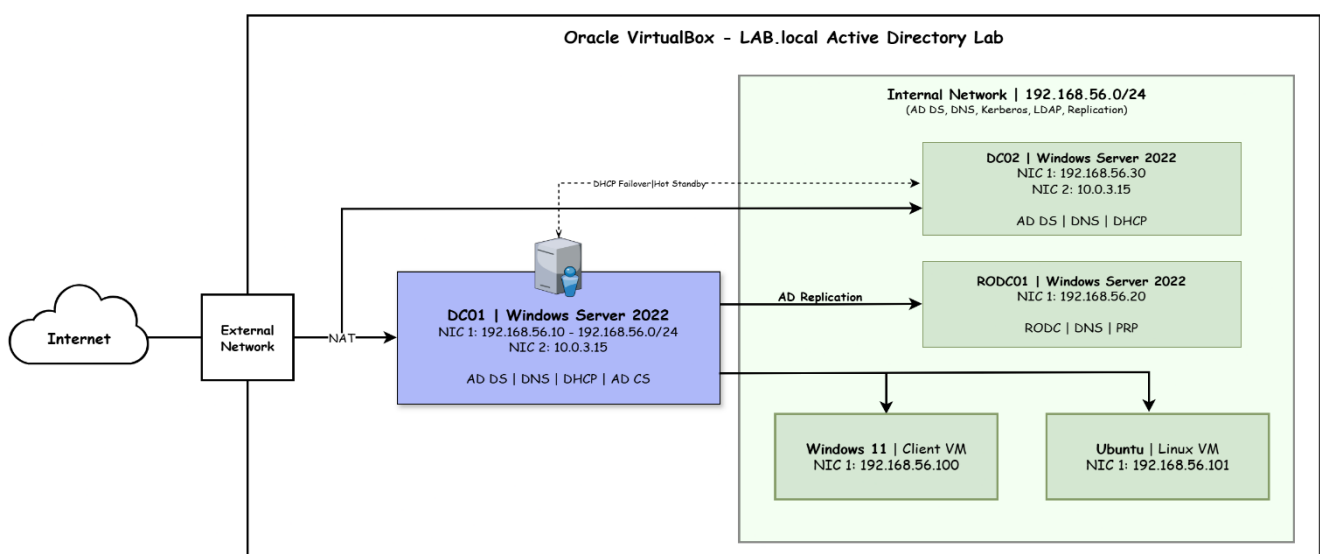


Figure 2.1.a - VirtualBox network topology showing domain controllers, client machines and network segmentation.

2.2. Domain and Naming

The lab uses a single Active Directory forest and domain, LAB.local. This naming convention is chosen to clearly indicate that the environment is a non-production, internal lab domain.

The domain controllers are named DC01 (Windows Server 2022), DC02 and RODC-Server, following a simple, role-based naming convention that makes it immediately clear which servers are writable domain controllers and which one is the Read-Only Domain Controller. Client machines follow descriptive names such as Windows11 for the first client workstation and Ubuntu-LAB for the Linux client.



Figure 2.2.a - VirtualBox Manager showing all running virtual machines in the lab.

3. Active Directory Design

This section describes the logical design of the Active Directory environment, including the domain controller topology, the FSMO role distribution, replication behavior, and the Organizational Unit (OU) structure used to scope Group Policy and delegate administrative permissions. The design reflects common enterprise practices adapted to a lab-scale environment, prioritizing clarity, redundancy, and least-privilege administration.

3.1. Domain Controllers

The domain LAB.local is served by three domain controllers, each fulfilling a distinct role within the environment.

DC01 acts as the primary domain controller and holds all five FSMO (Flexible Single Master Operation) roles: Schema Master, Domain Naming Master, RID Master, PDC Emulator, and Infrastructure Master. It is also configured as a Global Catalog (GC) server, enabling forest-wide search and universal group membership resolution. In addition, DC01 hosts DNS, DHCP, and Active Directory Certificate Services (AD CS).

DC02 is a secondary, fully writable domain controller that provides redundancy for authentication, DNS resolution, and DHCP services. It participates in multi-master replication with DC01, ensuring that a single point of failure does not interrupt domain operations.

RODC01 is a Read-Only Domain Controller, deployed to simulate a branch-office scenario where the physical security of the server cannot be fully guaranteed. It holds a read-only copy of the AD database and enforces a Password Replication Policy (PRP) to control which user credentials are cached locally.

A dedicated Password Settings Object was also configured for privileged accounts to enforce stricter password requirements independently of the domain-wide default policy.

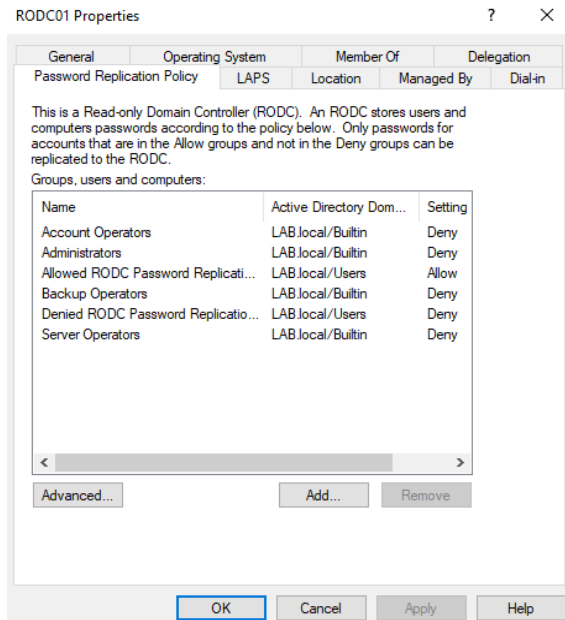


Figure 3.1.a - RODC01 Password Replication Policy showing the groups allowed and denied for password replication.

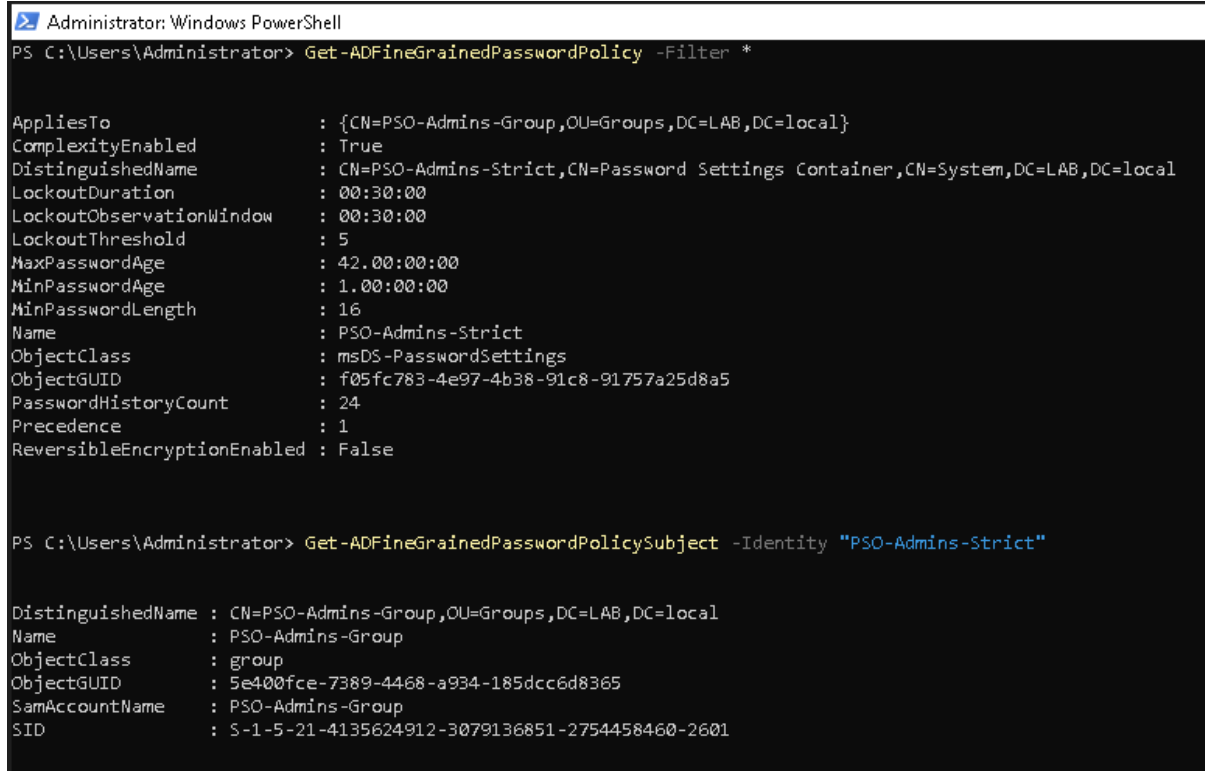


Figure 3.1.b - Get-ADFineGrainedPasswordPolicy confirming the PSO-Admins-Strict Password Settings Object, applied to PSO-Admins-Group with stricter password requirements (16-character minimum, 24-password history, and highest precedence).

Replication between all three domain controllers was verified using repadmin and dcdiag, as documented in Section 9 (Validation and Testing).

Design note - DC01 Network Configuration: DC01 is multi-homed, with two network adapters: an internal host-only adapter (192.168.56.10) used for domain traffic including replication, Kerberos authentication, and internal DNS resolution and a NAT adapter (10.0.3.x) used exclusively to provide the VM with internet access for system updates and package downloads during lab setup. Running Get-ADDomainController returns the NAT adapter's address rather than the internal one, since that interface most recently registered its IP with dynamic DNS, based on the configured network adapter binding order. This does not affect domain functionality, as replication, authentication, and internal name resolution continue to operate correctly over the internal interface, as verified with **repadmin** and **dcdiag** above. In a production environment, domain controllers should not be multi-homed, as this can introduce ambiguity in DNS registration, replication topology, and SRV records. In this lab, the configuration was retained for practical convenience to simplify internet access for the VM without relying on a shared NAT Network rather than as an oversight.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

Host Name . . . . . : DC01
Primary Dns Suffix . . . . . : LAB.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : Yes
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : LAB.local

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-C3-3A-E0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.56.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 
DNS Servers . . . . . : 192.168.56.10
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter #2
Physical Address. . . . . : 08-00-27-CF-16-0C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 10.0.3.15(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : mardi 15 septembre 2026 15:45:34
Lease Expires . . . . . : mercredi 16 septembre 2026 15:45:32
Default Gateway . . . . . : 10.0.3.2
DHCP Server . . . . . : 10.0.3.2
DNS Servers . . . . . : 192.168.56.10
NetBIOS over Tcpip. . . . . : Enabled
```

Figure 3.1.c - ipconfig /all on DC01, showing the static internal IP address used for domain traffic and the NAT adapter providing internet access.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADDomainController -Filter * | Select Name, IPv4Address, IsReadOnly

Name      IPv4Address  IsReadOnly
-----
DC01      10.0.3.15    False
DC02      192.168.56.30 False
RODC01    192.168.56.20 True

PS C:\Users\Administrator>
```

Figure 3.1.d - PowerShell output confirming domain controller roles and read-only status.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> netdom query fsmo
Schema master           DC01.LAB.local
Domain naming master    DC01.LAB.local
PDC                     DC01.LAB.local
RID pool manager        DC01.LAB.local
Infrastructure master   DC01.LAB.local
The command completed successfully.

PS C:\Users\Administrator>
```

Figure 3.1.e - FSMO role holder confirmation via netdom query fsmo.

3.2. OU Structure and Naming

The Organizational Unit (OU) structure of the LAB.local domain was designed following a separation-by-function and resource-type approach. Administrative accounts are isolated in the ADMINS OU, kept apart from standard user accounts hosted in USERS, applying the principle of least privilege and enabling differentiated Group Policy Object (GPO) assignment.

Security groups used to manage permissions and access are grouped under the Groups OU, while Linux-based machines in this case, the domain-joined Ubuntu host (PCLAB-UBUNTU) reside in a dedicated Linux OU, allowing Linux-specific policies to be applied without interfering with Windows clients.

By default, Active Directory places newly joined computer objects in the built-in Computers container. In this lab, the Windows 11 client machine's computer object was relocated to a custom OU named _COMPUTERS OU to demonstrate targeted computer-level GPO assignment. This change is reflected in Section 6.3, where the AppLocker Computer Configuration policy is linked to the _COMPUTERS OU and applied at startup, before any user logon, illustrating both the mechanics and the operational risks of computer-level GPO targeting. In a production environment, an equivalent effect can be achieved for all future computer joins using the redircmp command (redircmp "OU=Workstations, DC=LAB, DC=local"), which redirects newly joined computers to a specified OU by default, enabling consistent GPO application across all domain-joined devices without requiring manual relocation.

The Domain Controllers OU, automatically generated by Active Directory, hosts the domain controllers DC01, DC02, and RODC01. Finally, an additional OU named Extra was reserved for future use or testing purposes, and currently remains empty.

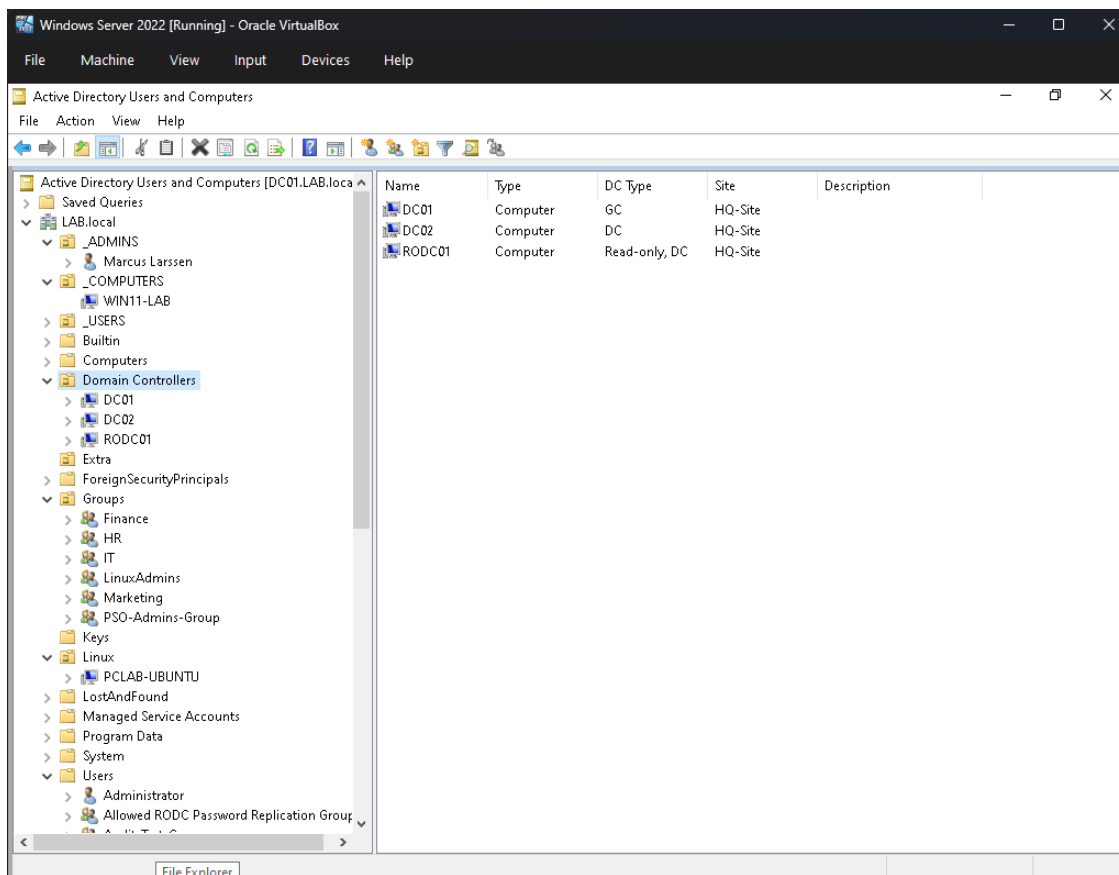


Figure 3.2.a - Active Directory Users and Computers showing the three domain controllers within LAB.local

4. Server Configuration

This section documents the specific server roles, features, and configuration details for each domain controller in the lab: DC01, DC02, and RODC01. While Section 3 described the logical design and topology, this section focuses on the concrete Windows Server roles and services installed on each machine, along with the configuration choices made to support the lab objectives.

4.1. DC01 – Primary Domain Controller

DC01 runs Windows Server 2022 Standard (Evaluation) and hosts the following server roles: Active Directory Domain Services (AD DS), DNS Server, DHCP Server, and Active Directory Certificate Services (AD CS). As the first domain controller promoted in the LAB.local forest, DC01 holds all five FSMO roles and operates as a Global Catalog server, as detailed in Section 3.1.

DC01 is configured with a static internal IP address (192.168.56.10) on its primary network adapter, used for all domain traffic, DNS resolution, and Kerberos authentication. A secondary NAT-connected adapter provides outbound internet access for system updates, as documented in Section 3.1 (Figure 3.1.c).

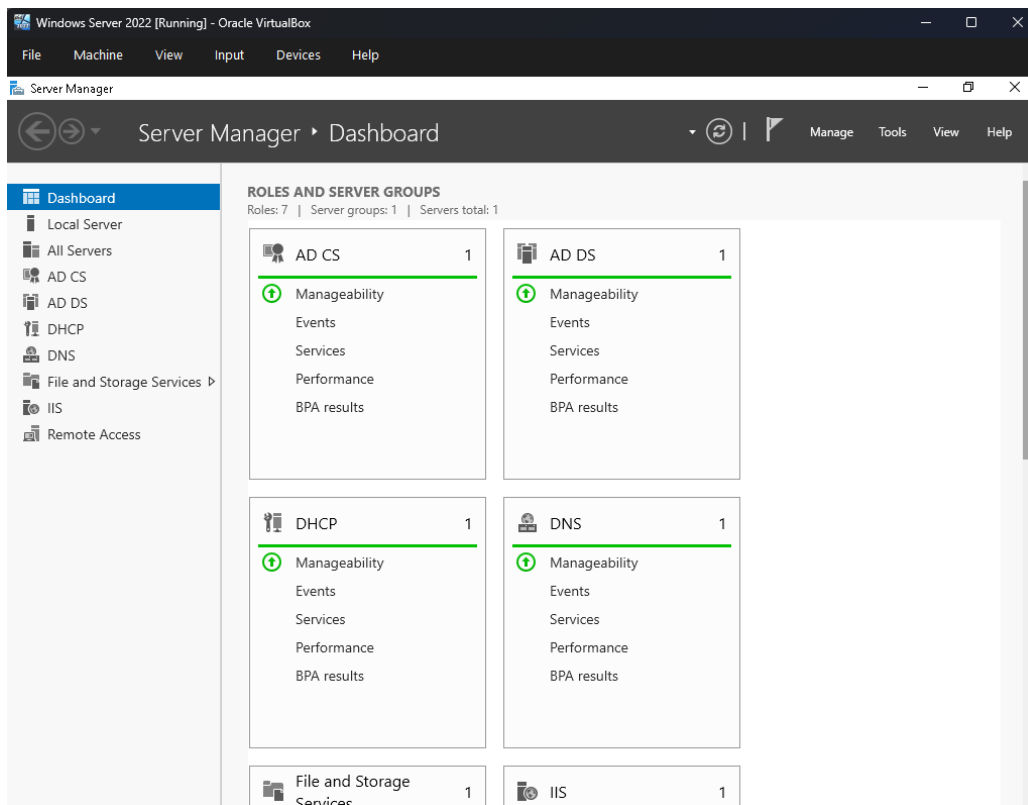


Figure 4.1.a – Server Manager dashboard on DC01, showing installed roles: AD DS, DNS Server, DHCP Server, and AD CS.

4.2. DC02 – Secondary Domain Controller

DC02 runs Windows Server 2022 Standard (Evaluation) and was promoted as an additional writable domain controller in the LAB.local domain, using DC01 as its replication source. It hosts the AD DS, DNS Server, and DHCP Server roles, providing redundancy for authentication, name resolution, and IP address assignment.

DC02 is configured with a static internal IP address (192.168.56.30). Its DHCP service is configured in Hot Standby failover mode with DC01, ensuring that DHCP leases continue to be issued to clients even if one of the two servers becomes unavailable.

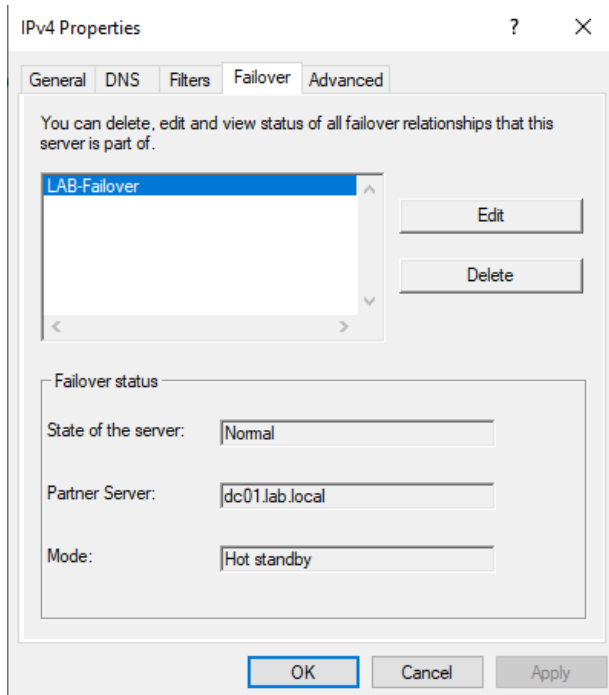


Figure 4.2.a – IPv4 Properties on DC02, Failover tab, showing the LAB-Failover relationship with DC01 in Hot Standby mode.

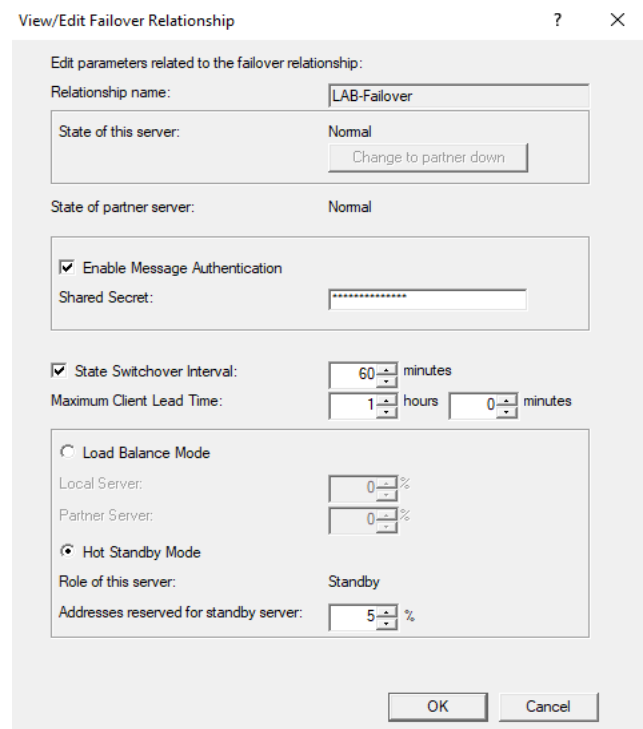


Figure 4.2.b – Detailed failover relationship parameters.

4.3. RODC01 – Read-Only Domain Controller

RODC01 runs Windows Server 2022 Standard (Evaluation) and was deployed as a Read-Only Domain Controller to simulate a branch-office scenario. It hosts a read-only copy of the Active Directory database and the DNS Server role, allowing local name resolution without exposing a writable copy of AD DS to a less physically secure location.

As detailed in Section 3.1, RODC01 enforces a Password Replication Policy (PRP) that restricts which user credentials can be cached locally, and it is excluded from FSMO role holding, consistent with its read-only nature confirmed in Figure 3.1.d.

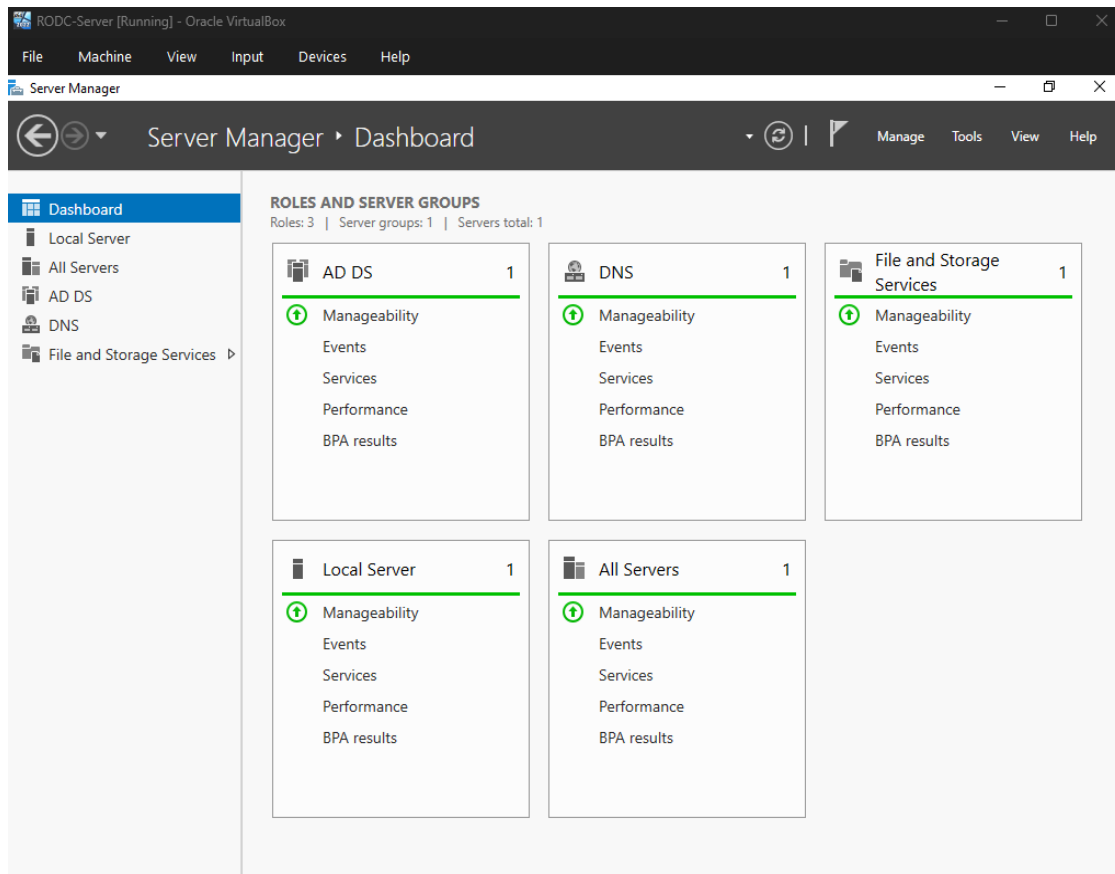


Figure 4.3.a – Server Manager dashboard on RODC01, confirming the AD DS and DNS Server roles.

5. Client Configuration

This section documents the configuration of the two client machines joined to the LAB.local domain: a Windows 11 workstation and an Ubuntu Linux host. Both clients obtain their network configuration via DHCP and authenticate against Active Directory, demonstrating cross-platform domain integration.

5.1. Windows 11 Domain Client

The Windows 11 client was joined to the LAB.local domain and receives its IP configuration dynamically from the DHCP service hosted on DC01/DC02. As shown in Section 3.2, the computer object is located in the _COMPUTERS OU.

At computer startup, the Computer Configuration policies linked to the domain and to the _COMPUTERS OU are applied, including AppLocker, Default Domain Policy, and Password Policies. Upon user logon, the User Configuration policies linked to the _USERS OU are applied as well, including the corporate wallpaper, Control Panel restrictions (Control Panel Block), folder redirection, and network drive mapping.

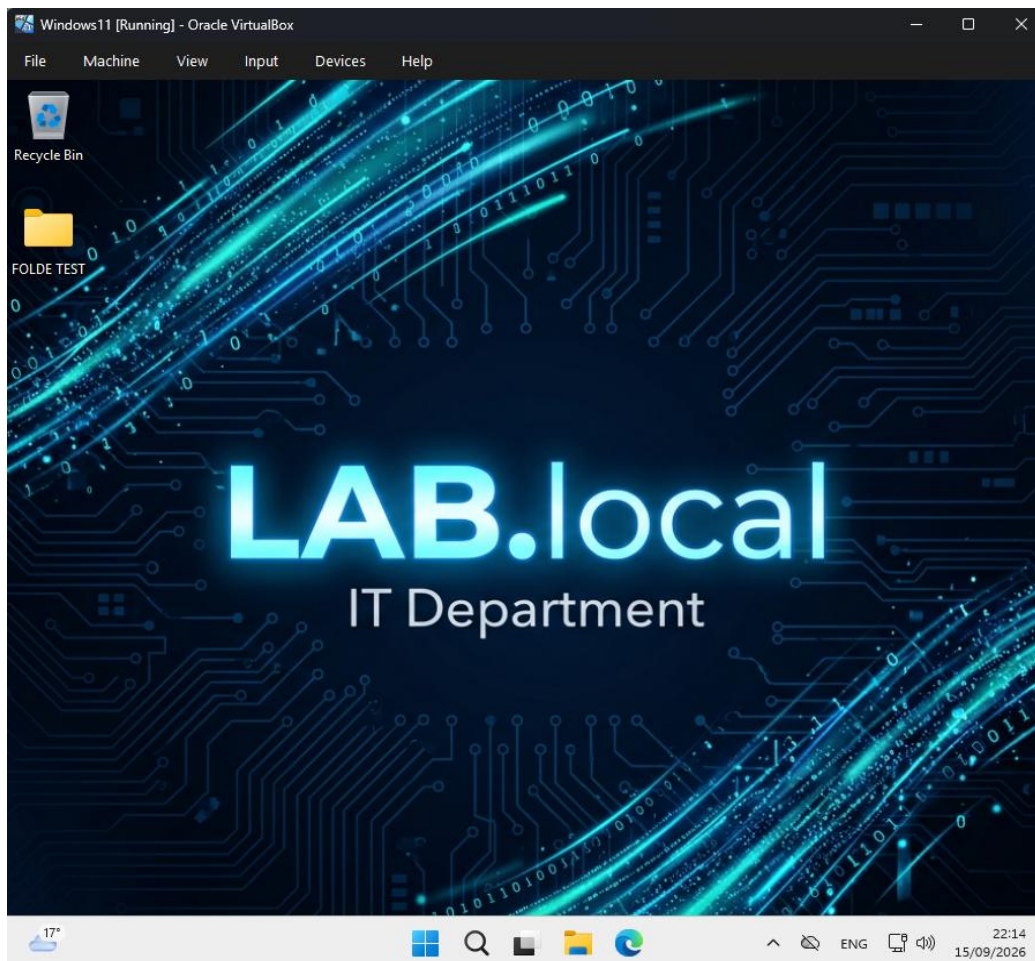


Figure 5.1.a - Windows 11 LAB client desktop showing the LAB.local corporate wallpaper.

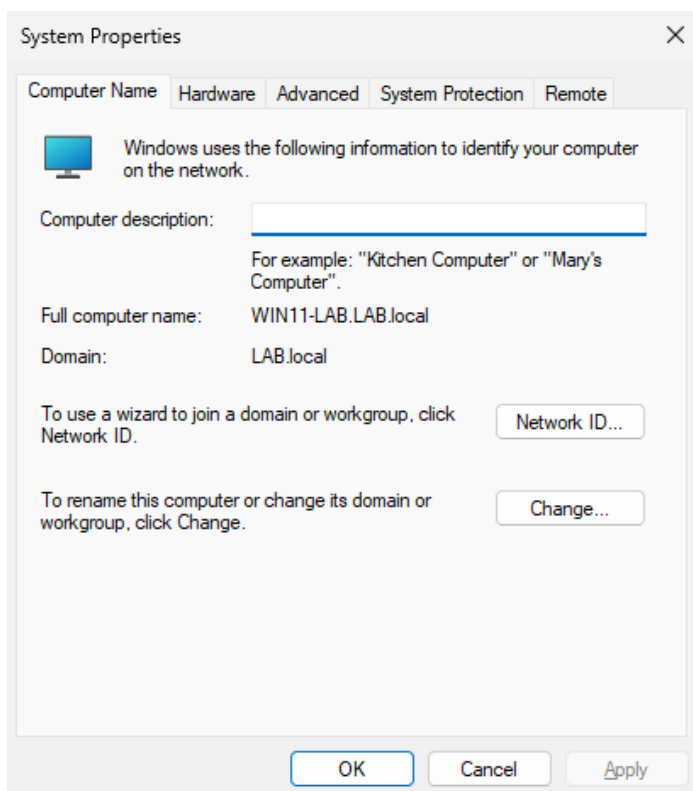


Figure 5.1.b – System Properties on the Windows 11 client, confirming domain membership in LAB.local.

```
PS C:\WINDOWS\system32> ipconfig /all

Windows IP Configuration

Host Name . . . . . : WIN11-LAB
Primary Dns Suffix . . . . . : LAB.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : LAB.local

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : LAB.local
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-A1-54-92
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::8e3f:41ea:57c:7b8a%12(Preferred)
IPv4 Address. . . . . : 192.168.56.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, 15 September 2026 22:05:30
Lease Expires . . . . . : Wednesday, 23 September 2026 22:05:30
Default Gateway . . . . . : 192.168.56.10
DHCP Server . . . . . : 192.168.56.10
DHCPv6 IAID . . . . . : 84410407
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-3D-1D-2E-08-00-27-A1-54-92
DNS Servers . . . . . : 192.168.56.10
NetBIOS over Tcpip. . . . . : Enabled
```

Figure 5.1.c - Network configuration of the WIN11-LAB client obtained via DHCP from the lab domain controllers.

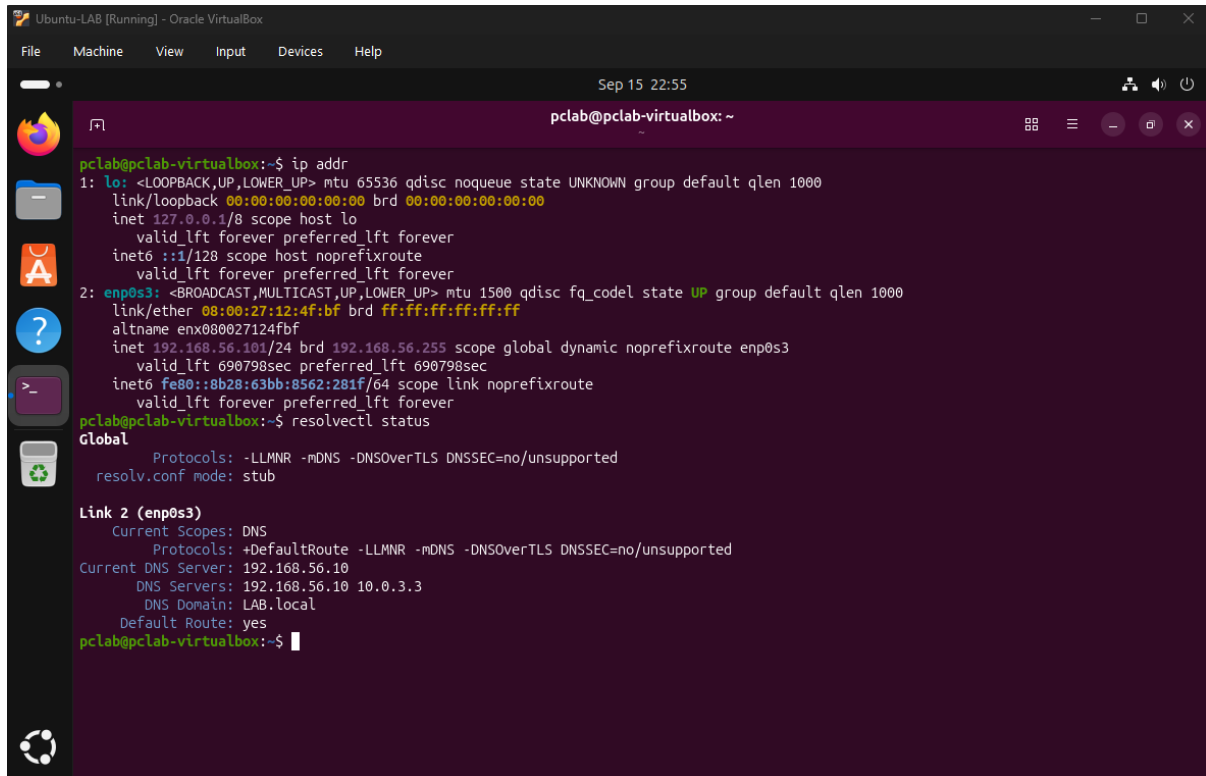
5.2. Ubuntu Client

The Ubuntu client was integrated into the LAB.local Active Directory domain using Samba and Winbind, allowing the Linux system to join the domain as a member server and authenticate users directly against Active Directory. This approach also enables the Ubuntu host to expose SMB shares that can be accessed using Active Directory credentials, demonstrating file-sharing interoperability between Linux and Windows in the lab environment.

Samba was configured through `/etc/samba/smb.conf`, defining the workgroup, realm and security settings required for Active Directory integration. Winbind was then used to handle the communication between the local system and the domain controllers, including user and group resolution via the Name Service Switch (NSS).

5.2.1. Network Configuration

Before configuring Samba, the Ubuntu client was set to use the internal Active Directory DNS service, since domain discovery and Kerberos authentication depend on correct name resolution. The client obtains its IP configuration via DHCP and uses the lab domain controllers as DNS servers.

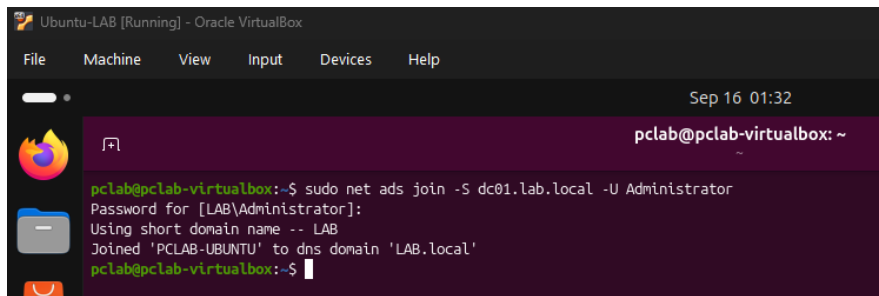


```
pcclab@pcclab-virtualbox: ~  
pcclab@pcclab-virtualbox:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:12:4f:bf brd ff:ff:ff:ff:ff:ff  
    altname enx080027124fbf  
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3  
        valid_lft 690798sec preferred_lft 690798sec  
    inet6 fe80::8b28:63bb:8562:281f/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
pcclab@pcclab-virtualbox:~$ resolvectl status  
Global  
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported  
    resolv.conf mode: stub  
  
Link 2 (enp0s3)  
    Current Scopes: DNS  
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported  
    Current DNS Server: 192.168.56.10  
    DNS Servers: 192.168.56.10 10.0.3.3  
    DNS Domain: LAB.local  
    Default Route: yes  
pcclab@pcclab-virtualbox:~$
```

Figure 5.2.1.a – Ubuntu network configuration showing DHCP-assigned IP addressing and DNS resolution through the LAB.local Active Directory environment.

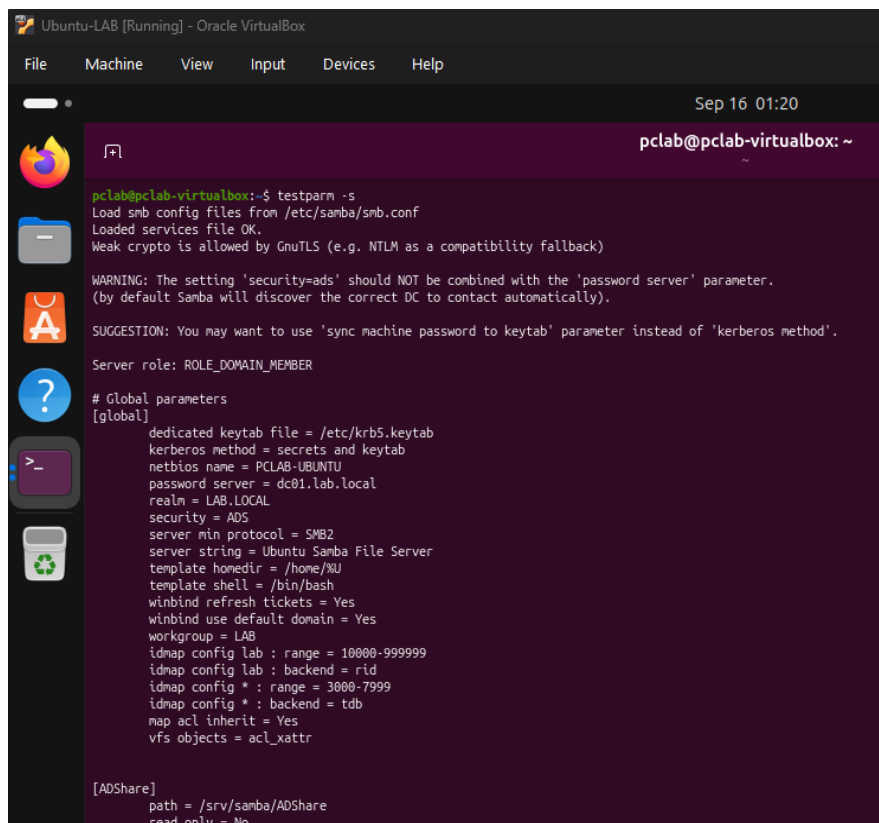
5.2.2. Samba Configuration and Domain Join

Samba was configured as an Active Directory domain member with security = ADS, realm = LAB.LOCAL and workgroup = LAB. User and group ID mapping was configured using the rid backend for the LAB domain (range 10000–999999) and the tdb backend for the default range (3000–7999), ensuring consistent and deterministic UID/GID assignment for Active Directory accounts. A dedicated file share, ADShare, was created at /srv/samba/ADShare to test file access using domain credentials.



```
Ubuntu-LAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 16 01:32
pclab@pclab-virtualbox: ~
pclab@pclab-virtualbox:~$ sudo net ads join -S dc01.lab.local -U Administrator
Password for [LAB\Administrator]:
Using short domain name -- LAB
Joined 'PCLAB-UBUNTU' to dns domain 'LAB.local'
pclab@pclab-virtualbox:~$
```

Figure 5.2.2.a - Output of net ads join, confirming successful domain join.



```
Ubuntu-LAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 16 01:20
pclab@pclab-virtualbox: ~
pclab@pclab-virtualbox:~$ testparm -s
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

WARNING: The setting 'security=ads' should NOT be combined with the 'password server' parameter.
(by default Samba will discover the correct DC to contact automatically).

SUGGESTION: You may want to use 'sync machine password to keytab' parameter instead of 'kerberos method'.

Server role: ROLE_DOMAIN_MEMBER

# Global parameters
[global]
    dedicated keytab file = /etc/krb5.keytab
    kerberos method = secrets and keytab
    netbios name = PCLAB-UBUNTU
    password server = dc01.lab.local
    realm = LAB.LOCAL
    security = ADS
    server min protocol = SMB2
    server string = Ubuntu Samba File Server
    template homedir = /home/%U
    template shell = /bin/bash
    winbind refresh tickets = Yes
    winbind use default domain = Yes
    workgroup = LAB
    idmap config lab : range = 10000-999999
    idmap config lab : backend = rid
    idmap config * : range = 3000-7999
    idmap config * : backend = tdb
    map acl inherit = Yes
    vfs objects = acl_xattr

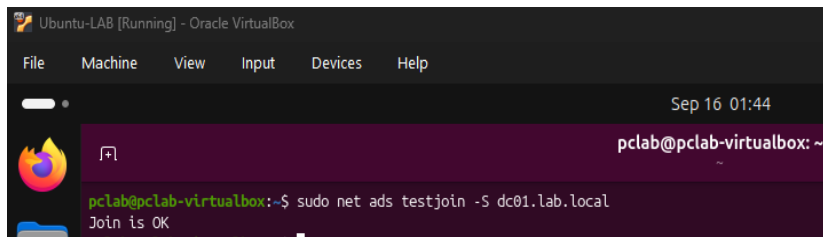
[ADShare]
    path = /srv/samba/ADShare
    read only = No
```

Figure 5.2.2.b - Output of testparm -s showing the loaded Samba global configuration.

5.2.3. Domain Join Verification

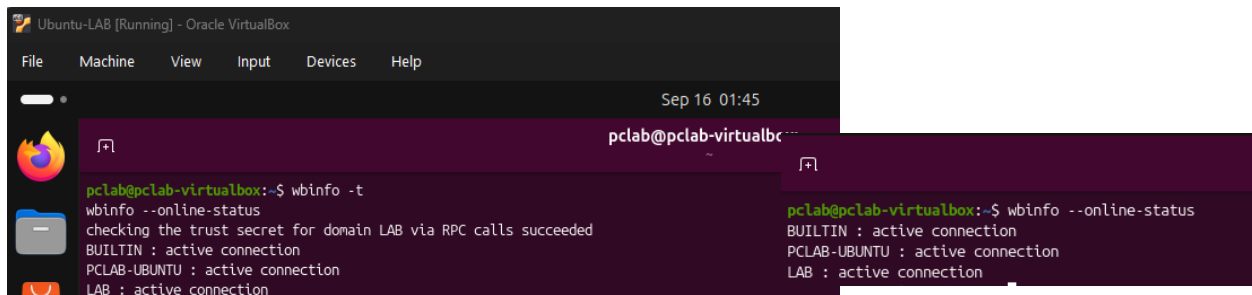
After completing the domain join, the membership was verified using `net ads testjoin`, which confirmed a valid Kerberos trust relationship between the Ubuntu client and the domain controller.

Winbind's connectivity to the domain was further validated using `wbinfo -t`, which checks the machine trust secret via RPC, which confirmed that the LAB domain was reachable and reporting an active connection. Together, these checks confirmed that the computer account was correctly established in Active Directory and that Winbind could communicate with the domain controller before proceeding to user-level authentication testing.



```
Ubuntu-LAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 16 01:44
pclab@pclab-virtualbox: ~
pclab@pclab-virtualbox:~$ sudo net ads testjoin -S dc01.lab.local
Join is OK
```

Figure 5.2.3.a - Output of `net ads testjoin`, confirming that the domain membership and Kerberos trust relationship are valid.



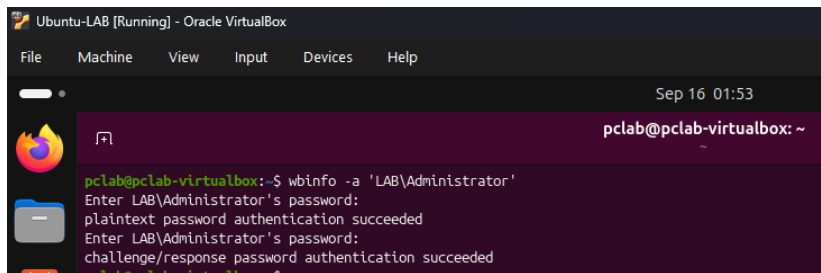
```
Ubuntu-LAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 16 01:45
pclab@pclab-virtualbox: ~
pclab@pclab-virtualbox:~$ wbinfo -t
wbinfo --online-status
checking the trust secret for domain LAB via RPC calls succeeded
BUILTIN : active connection
PCLAB-UBUNTU : active connection
LAB : active connection
pclab@pclab-virtualbox:~$ wbinfo --online-status
BUILTIN : active connection
PCLAB-UBUNTU : active connection
LAB : active connection
```

Figure 5.2.3.b - Output of `wbinfo -t` + `wbinfo --online-status`, confirming the machine trust secret is valid and Winbind maintains an active connection to the domain controller.

5.2.4. Authentication and Share Access Test

Domain user authentication was validated using `wbinfo -a`, which confirmed successful authentication against the domain controller using both plaintext and NTLM challenge-response methods. Interactive login was then tested using `su - Administrator`, which authenticated the domain account through PAM and automatically created a local home directory on first login, consistent with the template homedir setting configured in `smb.conf`.

To validate file-sharing interoperability, the ADShare file share was tested using smbclient. An initial attempt to write to the share using the Administrator account was correctly denied with NT_STATUS_ACCESS_DENIED, since the underlying directory permissions (POSIX mode 2770, owned by group linuxadmins) do not grant access to that account despite its elevated privileges within Active Directory. This confirmed that Samba enforces filesystem-level access control independently of AD role, following the principle of least privilege. The test was repeated using a domain account that is a member of the linuxadmins group, which successfully created a directory, uploaded a file, listed the share contents, and retrieved the file. A byte-for-byte comparison between the original and retrieved file confirmed data integrity across the SMB transfer.

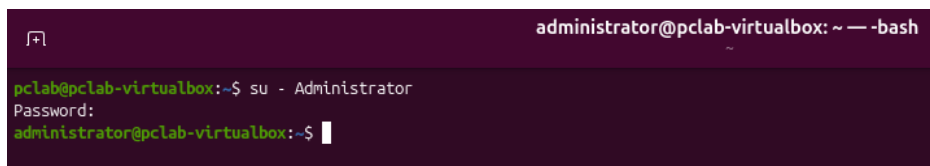


```

pclab@pclab-virtualbox:~$ wbinfo -a 'LAB\Administrator'
Enter LAB\Administrator's password:
plaintext password authentication succeeded
Enter LAB\Administrator's password:
challenge/response password authentication succeeded

```

Figure 5.2.4.a - Output of `wbinfo -a`, confirming successful plaintext and NTLM challenge-response authentication for a domain account.

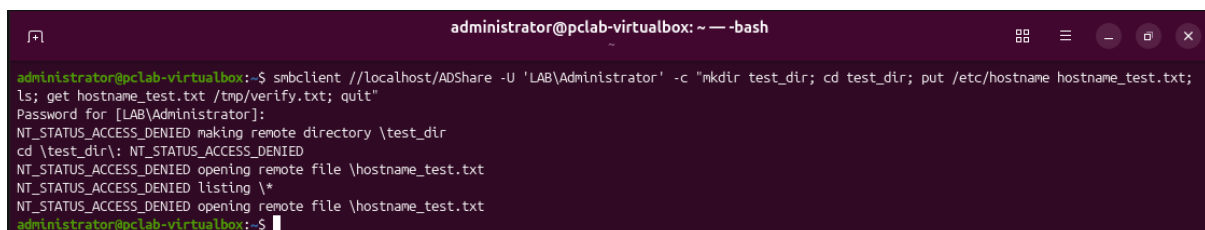


```

pclab@pclab-virtualbox:~$ su - Administrator
Password:
administrator@pclab-virtualbox:~$

```

Figure 5.2.4.b - Interactive login via `su -` as a domain account, showing automatic home directory creation by PAM upon first login.

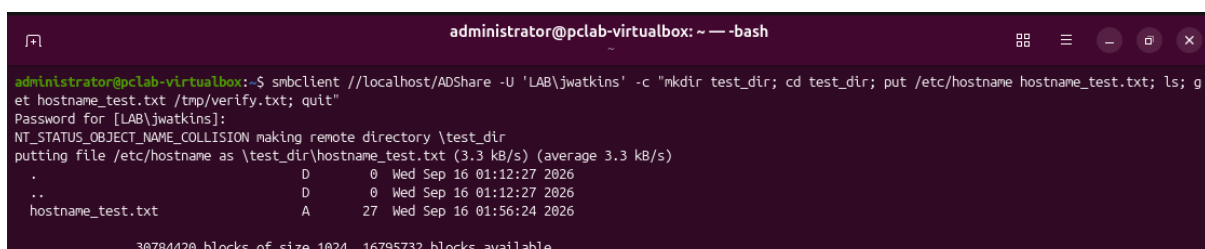


```

administrator@pclab-virtualbox:~$ smbclient //localhost/ADShare -U 'LAB\Administrator' -c "mkdir test_dir; cd test_dir; put /etc/hostname hostname_test.txt;
ls; get hostname_test.txt /tmp/verify.txt; quit"
Password for [LAB\Administrator]:
NT_STATUS_ACCESS_DENIED making remote directory \test_dir
cd \test_dir\; NT_STATUS_ACCESS_DENIED
NT_STATUS_ACCESS_DENIED opening remote file \hostname_test.txt
NT_STATUS_ACCESS_DENIED listing \*
NT_STATUS_ACCESS_DENIED opening remote file \hostname_test.txt
administrator@pclab-virtualbox:~$

```

Figure 5.2.4.c - Access attempt to the ADShare file share using the Administrator account, denied due to POSIX filesystem permissions restricting access to the linux admins group.



```

administrator@pclab-virtualbox:~$ smbclient //localhost/ADShare -U 'LAB\jwatkins' -c "mkdir test_dir; cd test_dir; put /etc/hostname hostname_test.txt; ls; g
et hostname_test.txt /tmp/verify.txt; quit"
Password for [LAB\jwatkins]:
NT_STATUS_OBJECT_NAME_COLLISION making remote directory \test_dir
putting file /etc/hostname as \test_dir\hostname_test.txt (3.3 kB/s) (average 3.3 kB/s)
.          D          0   Wed Sep 16 01:12:27 2026
..         D          0   Wed Sep 16 01:12:27 2026
hostname_test.txt  A      27   Wed Sep 16 01:56:24 2026

30784420 blocks of size 1024, 16795732 blocks available

```

Figure 5.2.4.d - Successful file share access test using a domain account belonging to the linux admins group.

These tests confirm that both Windows and Ubuntu clients can join the LAB.local domain, obtain network settings from DHCP, and authenticate against Active Directory using Kerberos and NTLM. Together, they demonstrate cross-platform integration of Linux and Windows in the same AD environment.

6. Group Policy Design

6.1. GPO Overview

Group Policy was used to centralize security settings, user restrictions and workstation configuration across the LAB.local domain. The policies were designed to separate domain-wide settings from settings targeted at domain controllers, users and client computers.

The lab contains nine custom GPOs and two default Active Directory policies. The custom policies cover application control, password security, auditing, BitLocker recovery, desktop standardization, folder redirection and network resource access.

Group Policy processing was tested on the Windows 11 client after the computer joined the domain. The resulting configuration was reviewed using `gpupdate /force` and `gpresult`, while the visible effects of selected policies were verified directly on the client.

6.2. GPO Structure and Scope

The Group Policy Objects were linked according to the type of object they manage. Domain-wide policies were linked at the LAB.local domain level, while computer-specific policies were linked to the appropriate organizational unit (OU). User-focused policies were linked to the OU containing the relevant user accounts.

This structure limits the scope of each policy and reduces the risk of applying server-specific settings to workstations or user-specific settings to domain controllers. The Default Domain Policy and Default Domain Controllers Policy remain available as baseline policies, while the custom GPOs provide additional configuration for the lab environment.

GPO	Scope	Main Purpose
Advanced Auditing	Domain Controllers OU	Enables security auditing on domain controllers
AppLocker	_COMPUTERS OU	Restricts execution of unauthorized applications
BitLocker Recovery	Domain Controllers OU	Stores BitLocker recovery information in Active Directory
Control Panel Block	_USERS OU	Prevents standard users from accessing Control Panel and PC settings
Corporate Wallpaper	_USERS OU	Applies a standard corporate desktop wallpaper
Folders Redirection	_USERS OU	Redirects selected user folders to a central network location
Map Network Drive	_USERS OU	Maps a shared network drive at user logon
Password Policies	Domain Level	Defines domain password and account lockout requirements
Restrict Software Installation	_USERS OU	Prevents standard users from installing unauthorized software

6.3. Group Policy Management Console

The GPO links were reviewed to confirm that each policy was associated with the intended domain, OU or computer group. This is particularly important for policies such as AppLocker, BitLocker Recovery and Advanced Auditing, because these settings are processed under Computer Configuration and must be linked to an OU containing computer objects.

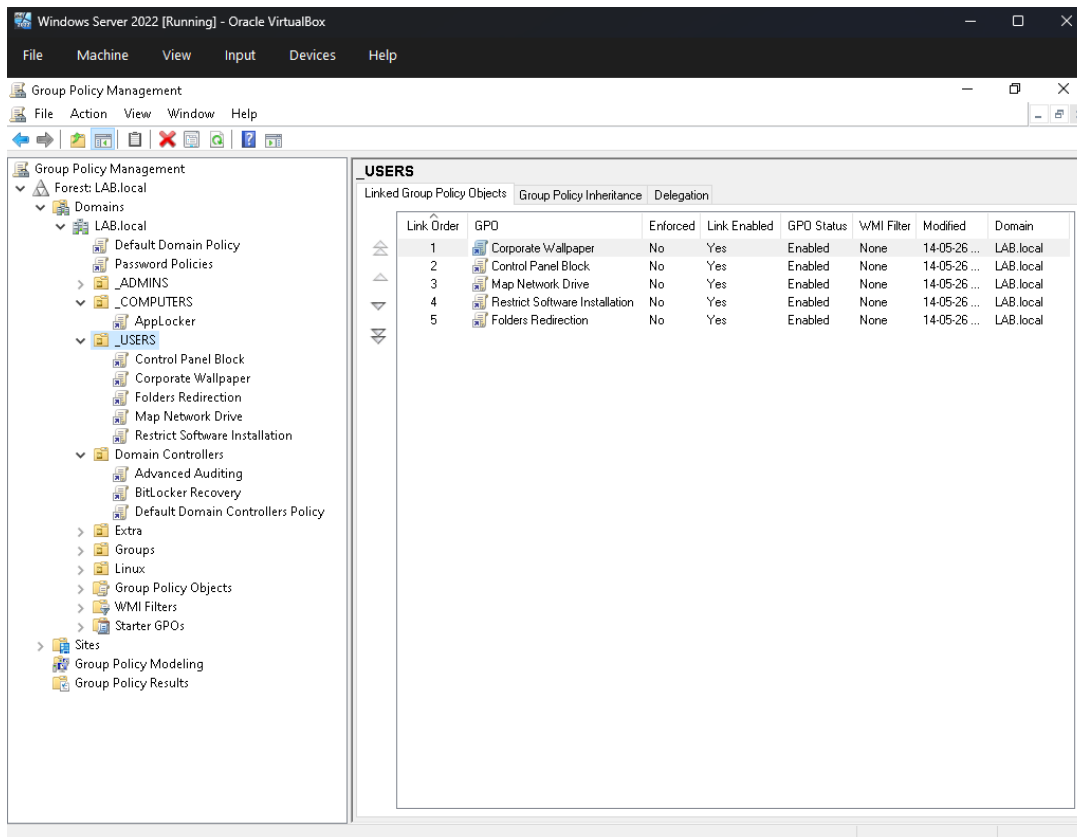


Figure 6.3.a - Group Policy Management Console displaying the LAB.local domain hierarchy, organizational units and linked Group Policy Objects.

6.4. Security and Hardening Policies

Several GPOs were implemented to strengthen the security posture of the lab environment. The Password Policies GPO defines password complexity, minimum password length, password expiration and account lockout requirements at the domain level.

The Advanced Auditing GPO enables auditing categories on domain controllers, including authentication events, account management and policy changes. AppLocker restricts the execution of unauthorized applications, while Restrict Software Installation prevents standard users from installing software without administrative approval.

BitLocker Recovery was configured to store recovery information in Active Directory for the targeted computer objects. This provides a centralized recovery mechanism and demonstrates how recovery data can be associated with domain computer accounts.

Design note - In addition to the default AppLocker rules, a custom rule was created to deny execution of Google Chrome (for testing) based on its file hash, demonstrating granular application control beyond the default Path and Publisher conditions.

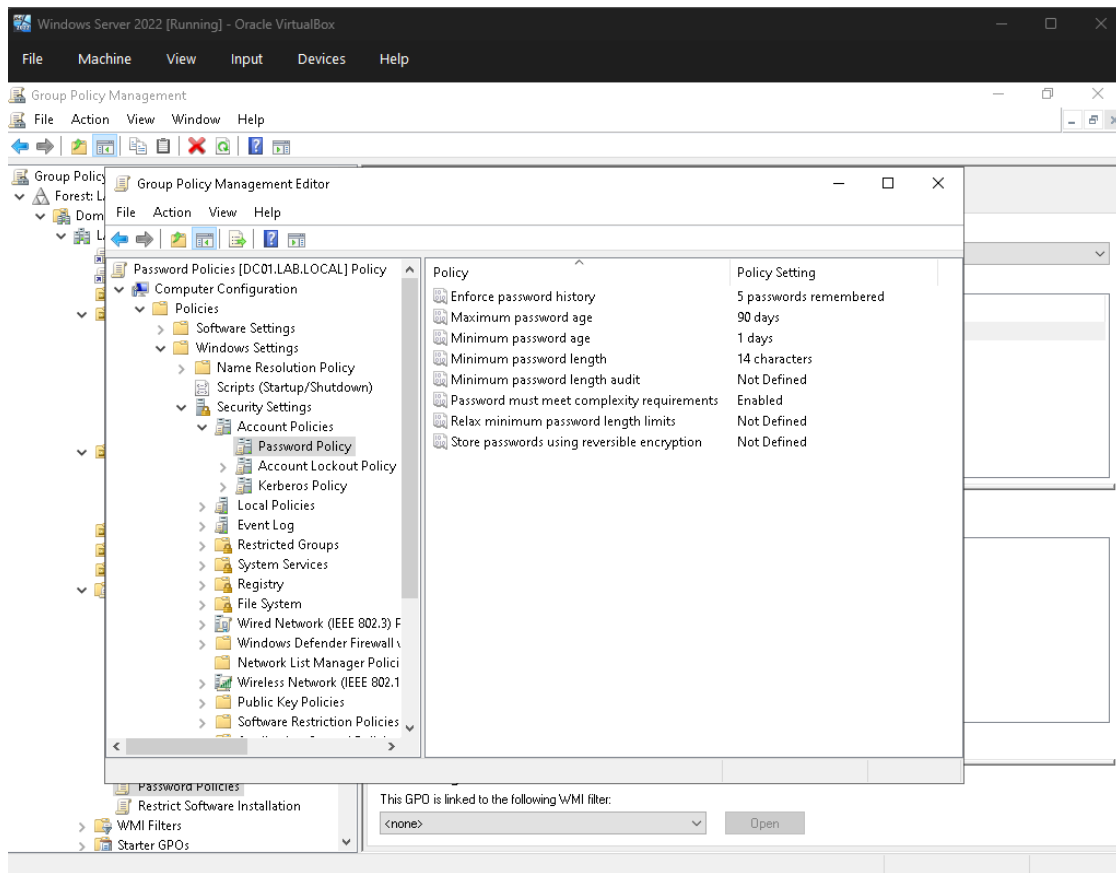


Figure 6.4.a - Password Policy settings configured for the LAB.local domain.

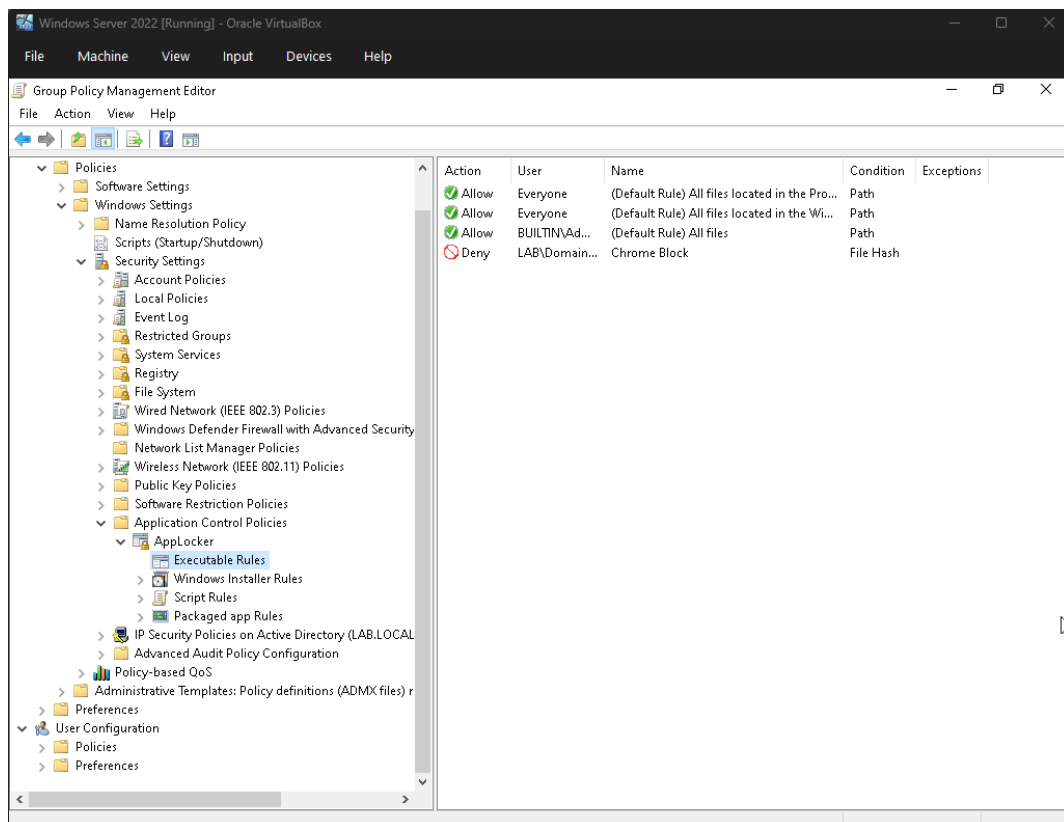


Figure 6.4.b - AppLocker executable rules configured through Group Policy to restrict unauthorized application execution.

6.5. User Experience & Resource Access Policies

Additional GPOs were created to standardize the user experience and simplify access to network resources. Corporate Wallpaper applies a consistent desktop background to domain users, while Control Panel Block restricts access to system configuration interfaces.

Folders Redirection moves selected user folders to a centralized network location, helping preserve user data independently of the local workstation. Map Network Drive automatically connects users to a shared network location during logon, providing consistent access to domain resources.



Figure 6.5.a - Corporate wallpaper applied to the Windows 11 client through Group Policy.

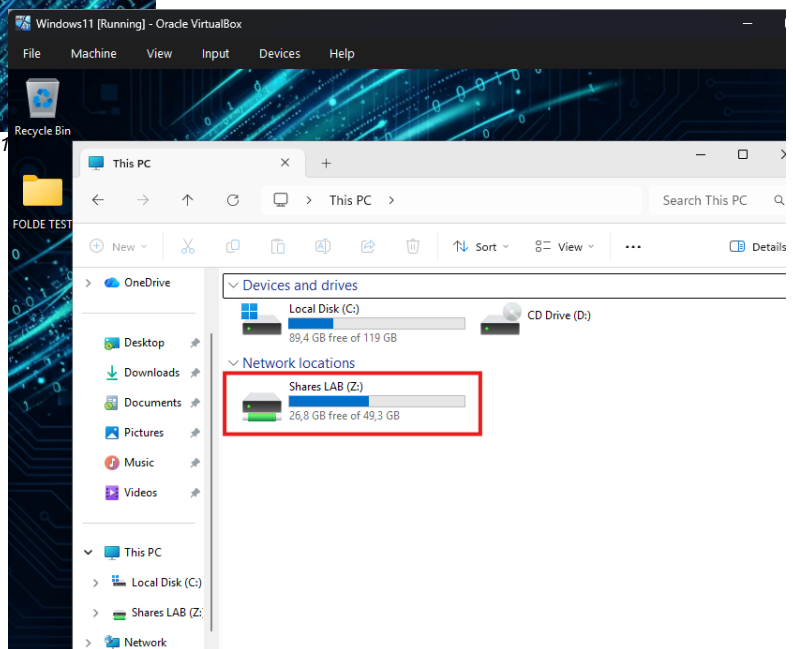


Figure 6.5.b - Network drive mapped automatically on the Windows 11 client after domain logon.

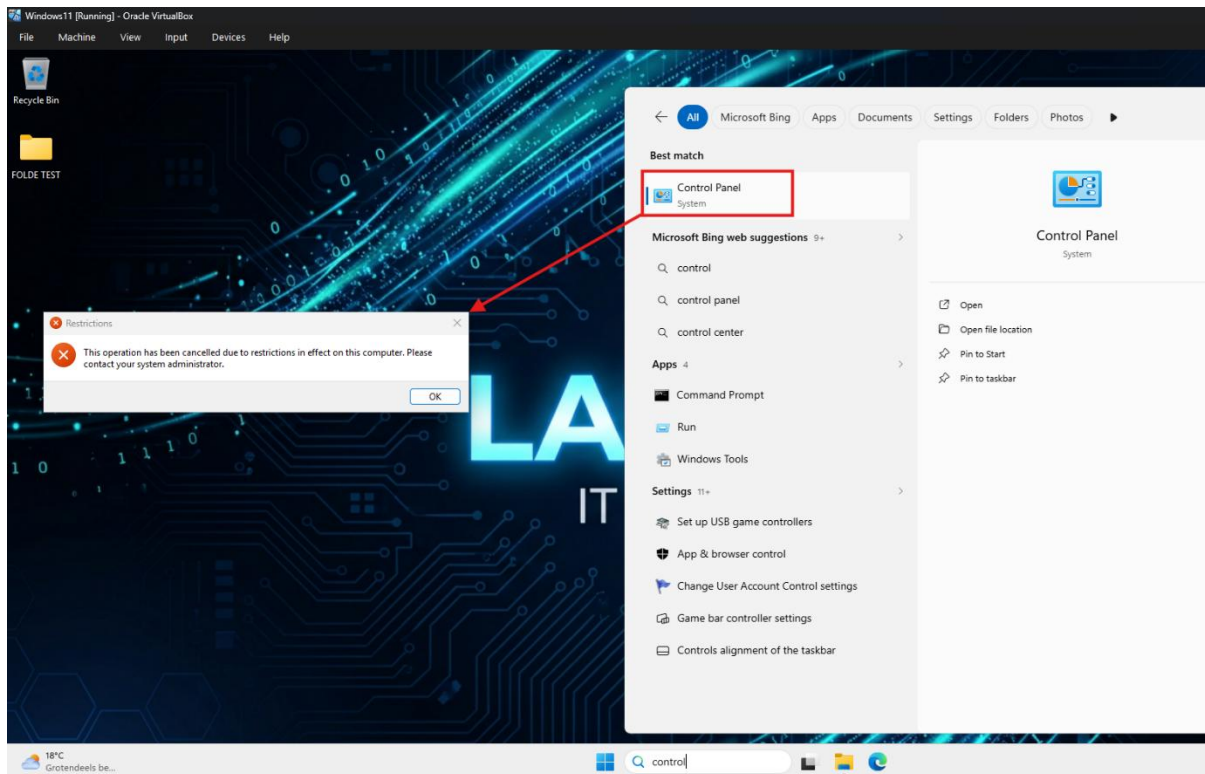


Figure 6.5.c – Control Panel access restricted on the Windows 11 client by Group Policy.

6.6. GPO Design Summary

The GPO design provides centralized management of security, user experience and workstation configuration settings, with each policy linked to the domain, an organizational unit or a computer group according to its intended scope. This separation ensures that domain-wide settings, domain controller settings, user settings and client computer settings are managed independently, reducing the risk of unintended policy application.

Computer Configuration settings, such as AppLocker rules and BitLocker recovery, are processed during system startup, while User Configuration settings, such as the corporate wallpaper and mapped network drive, are applied during user logon. This design demonstrates how Group Policy can be used to enforce security controls, standardize workstations and automate access to network resources in an Active Directory environment.

7. DNS and DHCP Configuration

7.1. DNS Zones and Records

The DNS service is hosted on DC01 and DC02, providing name resolution for the LAB.local Active Directory domain. Both domain controllers host the primary DNS zone, allowing clients to resolve internal hostnames regardless of which server responds to their query.

The DNS zone is Active Directory-integrated, meaning that zone data is replicated automatically between domain controllers as part of Active Directory replication. This removes the need for manual zone transfers and ensures that DNS records remain consistent across DC01, DC02 and RODC01.

Forward lookup records were verified to confirm that domain controllers and client machines can resolve each other by hostname. This is a prerequisite for Active Directory authentication, Group Policy processing and Kerberos ticket issuance, all of which rely on correct DNS resolution.

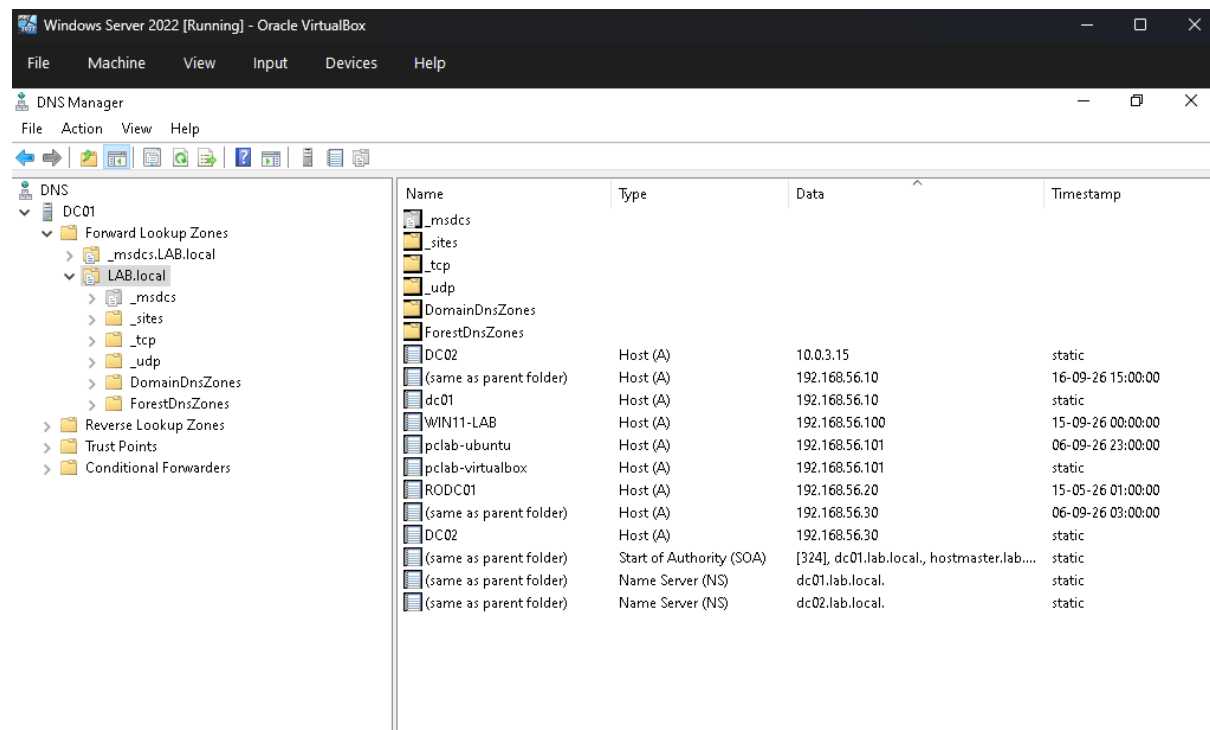


Figure 7.1.a – DNS Manager showing the LAB.local forward lookup zone and registered host records.

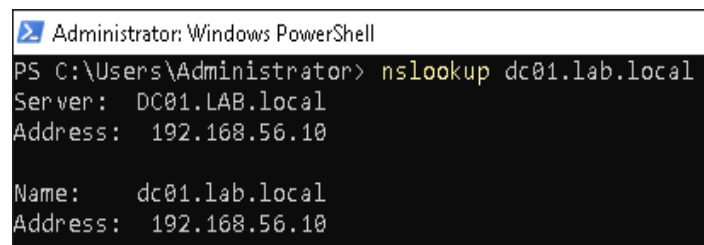


Figure 7.1.b – Forward DNS resolution of dc01.lab.local verified using nslookup.

7.2. DHCP Scopes and Failover

DHCP is configured on DC01 and DC02 to provide automatic IP addressing for client devices in the LAB.local network. Both servers share a single DHCP scope covering the 192.168.56.0/24 network, with DC01 acting as the primary DHCP server.

To provide high availability, DHCP Failover was configured between DC01 and DC02 in Hot Standby mode. In this configuration, DC01 operates as the active server under normal conditions, while DC02 remains on standby and automatically takes over lease distribution if DC01 becomes unavailable. This ensures that client devices can continue to obtain and renew IP addresses even if one of the two servers goes offline.

The DHCP scope includes standard options such as the default gateway, DNS server addresses and the domain name, ensuring that client devices are automatically configured to use the internal DNS infrastructure upon receiving an IP address.

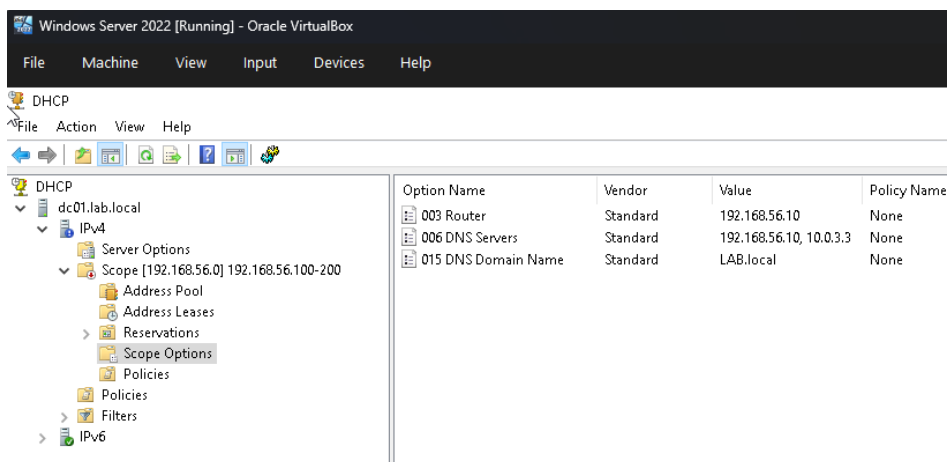


Figure 7.2.a - DHCP scope configuration on DC01 for the 192.168.56.0/24 network.

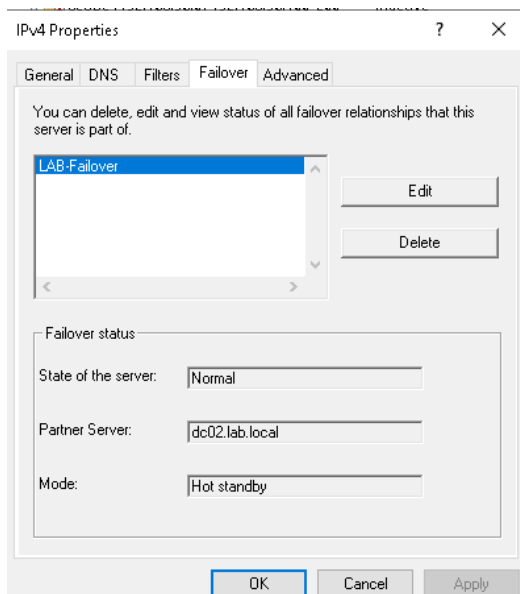


Figure 7.2.b - DHCP Failover configuration between DC01 and DC02 in Hot Standby mode.

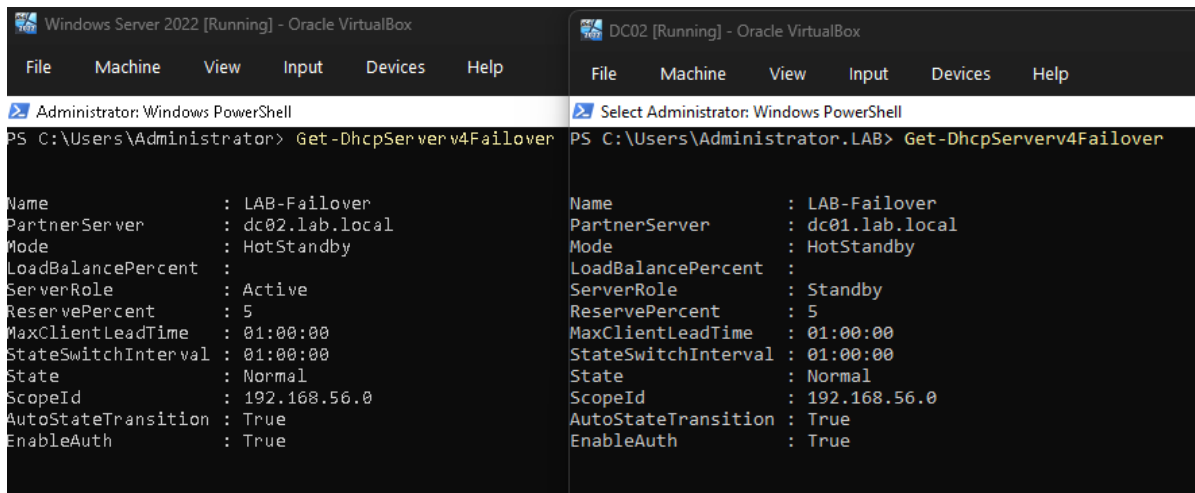


Figure 7.2.c - Get-DhcpServerv4Scope output on DC01 and DC02, confirming that the scope configuration is correctly synchronized between both failover partners.

8. Security Features

8.1. BitLocker Recovery

BitLocker Drive Encryption was enabled on the target domain controller to protect data at rest. Recovery is guaranteed through the BitLocker Recovery GPO (Section 6.4), linked to the Domain Controllers OU, which automatically backs up the recovery password and key package to Active Directory Domain Services (AD DS) for all domain controller computer objects.

This design avoids relying on a locally stored recovery key, which could be lost or inaccessible if the encrypted drive fails to boot. Instead, the recovery information is centrally stored in Active Directory and can be retrieved directly from the computer object properties by an authorized administrator.

The following evidence confirms that the recovery information was successfully backed up to Active Directory after BitLocker encryption was enabled.

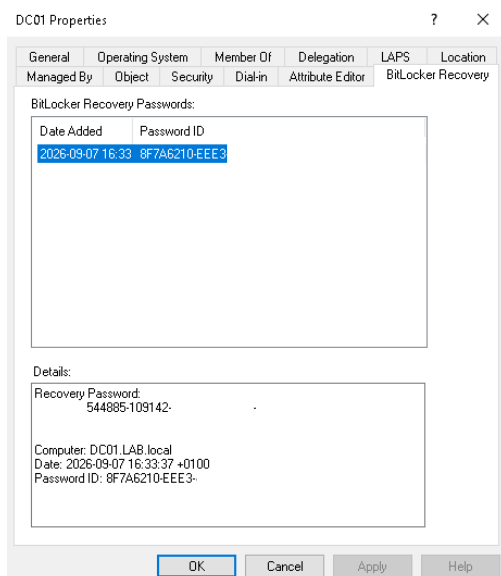


Figure 8.1.a - BitLocker recovery information stored in Active Directory for the target computer object.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-ADObject -Filter {objectclass -eq "msFVE-RecoveryInformation"} -SearchBase "CN=DC01,OU=Domain Controllers,DC=lab,DC=local" -Properties msFVE-RecoveryPassword, msFVE-RecoveryGuid, whenCreated | Format-List

DistinguishedName      : CN=2026-09-07T16:33:37+01:00{8F7A6210-EEE3-423F-8FC4-1B1759444B29},CN=DC01,OU=Domain
                        : Controllers,DC=LAB,DC=local
msFVE-RecoveryGuid      : {16, 98, 122, 143...}
msFVE-RecoveryPassword  : 544885-109142-
Name                   : 2026-09-07T16:33:37+01:00{8F7A6210-EEE3-423F-8FC4-1B1759444B29}
ObjectClass             : msFVE-RecoveryInformation
ObjectGUID             : 9344a453-15c6-4c29-ae7c-87b6e6e56fc9
whenCreated             : 07-09-26 17:01:53
```

Figure 8.1.b - PowerShell output confirming the BitLocker recovery password stored as an Active Directory object attribute.

8.2. AppLocker

While Section 6.4 presented the AppLocker rule configuration, including the default rules and a custom rule blocking Google Chrome by file hash, this section demonstrates the practical enforcement of that policy on the Windows 11 client.

After the AppLocker GPO was applied through gpupdate /force, an attempt was made to launch Google Chrome from the client machine. The application was blocked, confirming that the custom deny rule was correctly enforced and that AppLocker was actively preventing execution of unauthorized software, rather than only being configured at the policy level.

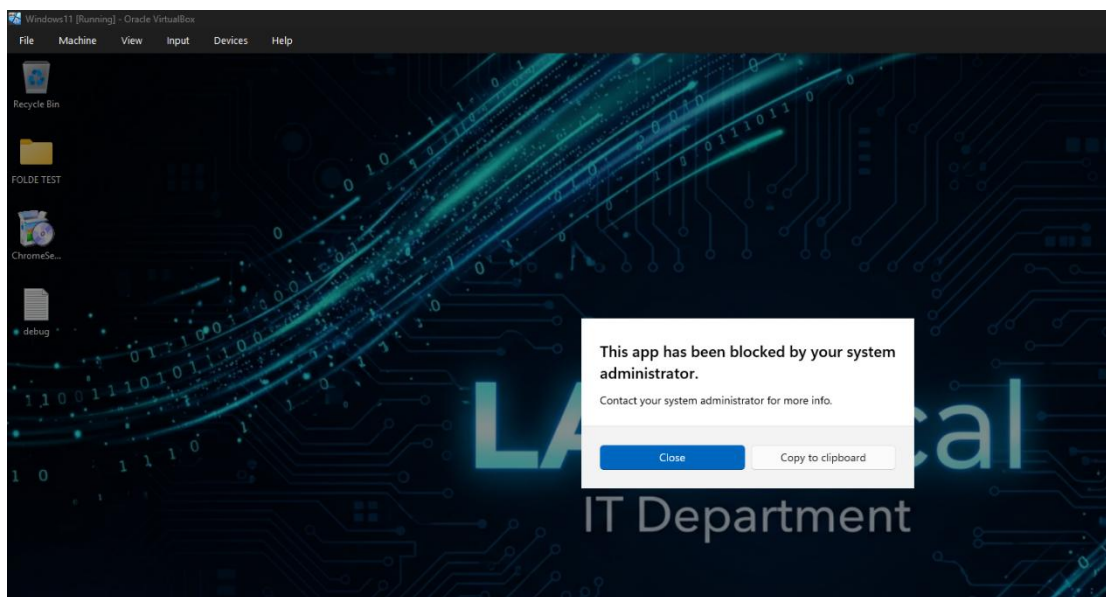


Figure 8.2.a - Google Chrome execution blocked on the Windows 11 client, confirming AppLocker policy enforcement.

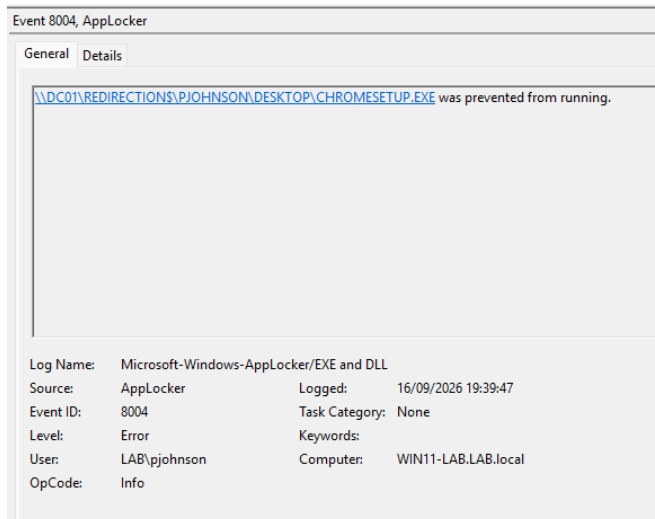


Figure 8.2.b - AppLocker Event Viewer log showing Event ID 8004, confirming that the Chrome execution attempt was blocked according to the configured policy.

8.3. Security Features Summary

These security features demonstrate two complementary layers of protection in the lab environment: data protection through centralized BitLocker recovery in Active Directory, and application control through AppLocker enforcement on the client. Together, they show that the policies configured in Section 6 are not only correctly defined, but also actively enforced and verifiable through practical testing.

9. Validation and Testing

This section documents the validation performed across the LAB.local environment to confirm that domain services, replication, client integration, Group Policy enforcement, and security features function as designed. Tests were performed using native Windows and Linux command-line tools, with results verified directly on the affected systems.

```

C:\Users\pjohnson>ipconfig /all

Windows IP Configuration

Host Name . . . . . : WIN11-LAB
Primary Dns Suffix . . . . . : LAB.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : LAB.local

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : LAB.local
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-A1-54-92
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::8e3f:41ea:57c:7b8a%12(Preferred)
IPv4 Address. . . . . : 192.168.56.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, 16 September 2026 19:32:31
Lease Expires . . . . . : Thursday, 24 September 2026 19:32:31
Default Gateway . . . . . : 192.168.56.10
DHCP Server . . . . . : 192.168.56.10
DHCPv6 IAID . . . . . : 84410407
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-3D-1D-2E-08-00-27-A1-54-92
DNS Servers . . . . . : 192.168.56.10
NetBIOS over Tcpip. . . . . : Enabled

```

Figure 9.a - ipconfig /all output on the Windows 11 client confirming a valid DHCP-assigned IP address (192.168.56.1xx), default gateway, and DNS server configuration.

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> repadmin /replsummary
Replication Summary Start Time: 2026-09-16 21:58:18

Beginning data collection for replication summary, this may take awhile:
.....

Source DSA          largest delta    fails/total %%   error
DC01                11m:42s        0 / 10          0
DC02                02m:45s        0 / 10          0

Destination DSA     largest delta    fails/total %%   error
DC01                02m:46s        0 / 5           0
DC02                11m:43s        0 / 5           0
RODC01             01m:49s        0 / 10          0

```

Figure 9.b - repadmin /replsummary output confirming successful Active Directory replication across DC01, DC02, and RODC01 with no errors.

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> netdom query fsmo
Schema master           DC01.LAB.local
Domain naming master    DC01.LAB.local
PDC                    DC01.LAB.local
RID pool manager        DC01.LAB.local
Infrastructure master   DC01.LAB.local
The command completed successfully.

```

Figure 9.c - netdom query fsmo output confirming that DC01 holds all five FSMO roles (Schema Master, Domain Naming Master, RID Master, PDC Emulator, and Infrastructure Master).

```
Command Prompt

C:\Users\pjohnson>gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 16/09/2026 at 22:00:37

RSOP data for LAB\pjohnson on WIN11-LAB : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.26200
Site Name:             N/A
Roaming Profile:        \\LAB.local\Profile$\pjohnson.V6
Local Profile:          C:\Users\pjohnson
Connected over a slow link?: No

USER SETTINGS
-----

CN=Patricia Johnson,OU=_USERS,DC=LAB,DC=local
Last time Group Policy was applied: 16/09/2026 at 21:21:47
Group Policy was applied from:      DC01.LAB.local
Group Policy slow link threshold:   500 kbps
Domain Name:                       LAB
Domain Type:                       Windows 2008 or later read-only DC

Applied Group Policy Objects
-----

Corporate Wallpaper
Control Panel Block
Map Network Drive
Folders Redirection

The following GPOs were not applied because they were filtered out
-----

Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----

Domain Users
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
LinuxAdmins
Authentication authority asserted identity
Medium Mandatory Level
```

Figure 9.d - gpresult /r output on the Windows 11 client confirming correct application of Computer Configuration and User Configuration Group Policy Objects.

```
Administrator: Windows PowerShell

PS C:\Users\Administrator.LAB> Get-DhcpServerv4Scope -ScopeId 192.168.56.0

ScopeId      SubnetMask    Name          State      StartRange    EndRange      LeaseDuration
-----
192.168.56.0 255.255.255.0 192.168.56.... Active      192.168.56.100 192.168.56.200 8.00:00:00

PS C:\Users\Administrator.LAB>
```

Figure 9.e - Get-DhcpServerv4Scope output confirming the 192.168.56.0 scope is active and correctly synchronized between DC01 and DC02.

```

pclab@pclab-virtualbox:~$ wbinfo -a jwatkins
Enter jwatkins's password:
plaintext password authentication succeeded
Enter jwatkins's password:
challenge/response password authentication succeeded

```

Figure 9.f - `wbinfo -a` output on the Ubuntu client confirming successful cross-platform authentication of a domain account against Active Directory via Winbind.

```

PS C:\Users\Administrator> nslookup dc01.lab.local
Server: DC01.LAB.local
Address: 192.168.56.10

Name:    dc01.lab.local
Address: 192.168.56.10

```

Figure 9.g - `nslookup` output confirming correct forward DNS resolution for `dc01.lab.local` within the `LAB.local` Active Directory-integrated zone.

```

PS C:\Users\Administrator> Get-ADUser pjohnson -Properties LockedOut

DistinguishedName : CN=Patricia Johnson,OU=_USERS,DC=LAB,DC=local
Enabled            : True
GivenName         : Patricia
LockedOut          : True
Name              : Patricia Johnson
ObjectClass       : user
ObjectGUID        : 84c81f3b-7076-4aef-b331-f466b9c39255
SamAccountName    : pjohnson
SID               : S-1-5-21-4135624912-3079136851-2754458460-1109
Surname           : Johnson
UserPrincipalName : pjohnson@LAB.local

```

Figure 9.h - `Get-ADUser` output confirming that a domain account was automatically locked out after exceeding the configured failed logon attempt threshold.

```

PS C:\Users\Administrator> repadmin /showrepl RODC01
HQ-Site\RODC01
DSA Options: IS_RODC
Site Options: (none)
DSA object GUID: fb2c5f36-89d7-443b-a265-549ef2f92d95
DSA invocationID: 833261da-37bf-4f57-a3b8-79549e3ee923

==== INBOUND NEIGHBORS =====
DC=LAB,DC=local
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 19:07:15 was successful.
HQ-Site\DC01 via RPC
DSA object GUID: bf79194a-8ded-4d7a-b594-2065f3f96bc7
Last attempt @ 2026-09-15 19:07:52 was successful.
CN=Configuration,DC=LAB,DC=local
HQ-Site\DC02 via RPC
DSA object GUID: bf79194a-8ded-4d7a-b594-2065f3f96bc7
Last attempt @ 2026-09-15 19:06:46 was successful.
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 19:06:46 was successful.
CN=Schema,CN=Configuration,DC=LAB,DC=local
HQ-Site\DC01 via RPC
DSA object GUID: bf79194a-8ded-4d7a-b594-2065f3f96bc7
Last attempt @ 2026-09-15 19:06:46 was successful.
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 19:06:46 was successful.
DC=DomainDnsZones,DC=LAB,DC=local
HQ-Site\DC01 via RPC
DSA object GUID: bf79194a-8ded-4d7a-b594-2065f3f96bc7
Last attempt @ 2026-09-15 19:06:46 was successful.
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 19:06:46 was successful.
DC=ForestDnsZones,DC=LAB,DC=local
HQ-Site\DC01 via RPC
DSA object GUID: bf79194a-8ded-4d7a-b594-2065f3f96bc7

```

```

PS C:\Users\Administrator> repadmin /showrepl DC01
HQ-Site\DC01
DSA Options: IS_GC
Site Options: (none)
DSA object GUID: bf79194a-8ded-4d7a-b594-2065f3f96bc7
DSA invocationID: bf79194a-8ded-4d7a-b594-2065f3f96bc7

==== INBOUND NEIGHBORS =====
DC=LAB,DC=local
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 15:52:32 was successful.
CN=Configuration,DC=LAB,DC=local
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 15:52:35 was successful.
CN=Schema,CN=Configuration,DC=LAB,DC=local
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 16:31:29 was successful.
DC=DomainDnsZones,DC=LAB,DC=local
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 16:29:41 was successful.
DC=ForestDnsZones,DC=LAB,DC=local
HQ-Site\DC02 via RPC
DSA object GUID: 0e2b1251-ba05-4aa7-a060-a3eaed627a60
Last attempt @ 2026-09-15 16:31:30 was successful.

```

Figure 9.i - Successful inbound replication results from RODC01 and DC01 using `repadmin /showrepl`.

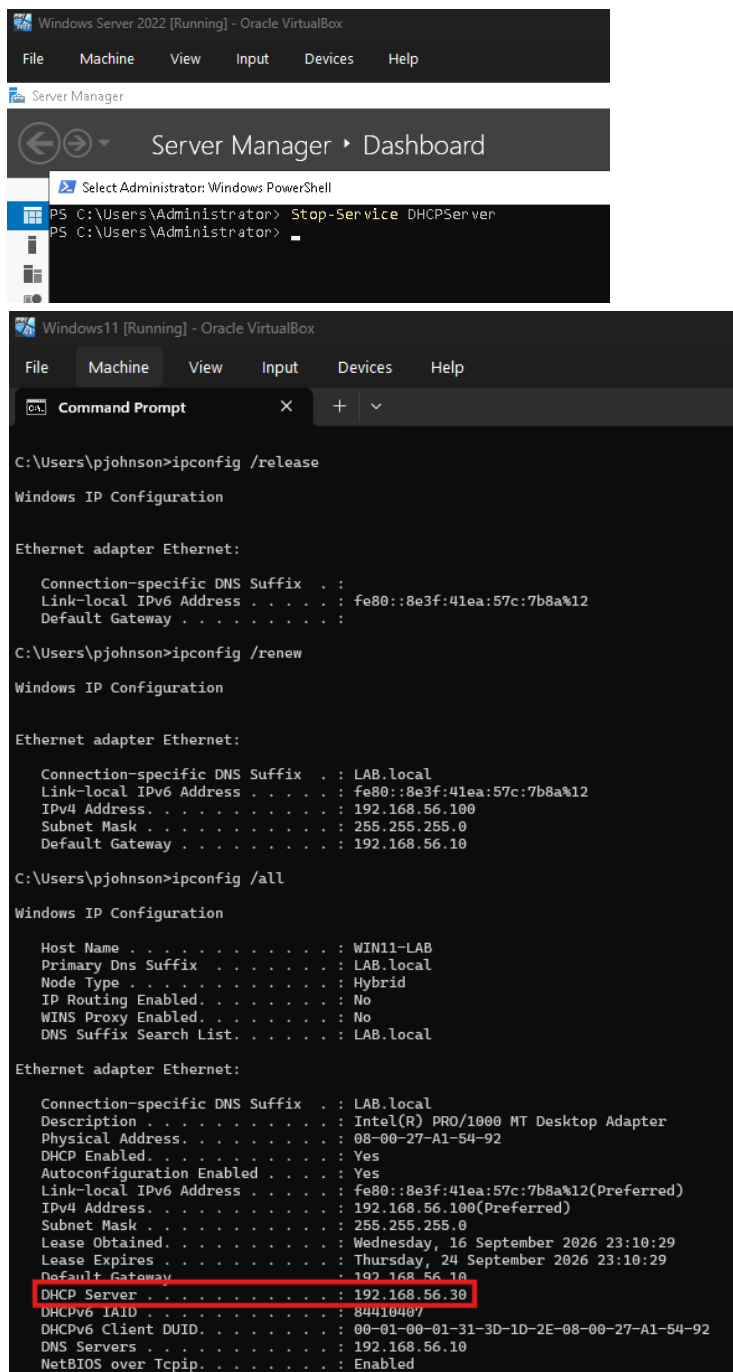


Figure 9.j - `ipconfig /all` output on the Windows 11 client confirming a lease obtained from DC02 (192.168.56.30) after the DHCP Server service was stopped on DC01, validating DHCP failover functionality.

10. Diagnostic Findings and Design Considerations

This section documents the diagnostic process behind two notable behaviors identified while validating the LAB.local environment and outlines potential improvements for extending the lab further. Rather than presenting a simple pass/fail record, it focuses on the reasoning and tools used to isolate root causes under realistic, production-like conditions.

1. DHCP Scope State as a Silent Failure Mode

As part of a broader review of the DHCP failover configuration between DC01 and DC02, the DHCP scope state was checked using `Get-DhcpServerv4Scope`. This revealed that the 192.168.56.0 scope on DC01 was configured as Inactive.

This condition is significant because the DHCP Server service can remain in a Running state while an individual scope is inactive. The scope range, subnet mask and failover relationship may still appear correctly configured, but the server will not assign leases from an inactive scope. On the client, the condition appeared as an APIPA address in the 169.254.0.0/16 range and a timeout when running `ipconfig /renew`.

The issue was isolated by checking the client configuration, DHCP service status, scope configuration and scope state separately. The scope was reactivated with the following command:

```
Set-DhcpServerv4Scope -ScopeId 192.168.56.0 -State Active
```

The client then renewed its lease successfully and received a valid address from the 192.168.56.0/24 network. This result was verified with `ipconfig /all`, confirming that the client had valid IPv4, DNS and DHCP server information.

2. Incremental Enforcement as an AppLocker Deployment Requirement

To evaluate AppLocker behavior under enforcement conditions, the four AppLocker rule collections. Executable, Windows Installer, Script and Packaged app were changed from Audit only to Enforce rules.

This test demonstrated why AppLocker enforcement should be introduced incrementally. The Packaged app Rules collection did not contain default allow rules when enforcement was enabled. Consequently, Windows 11 packaged applications, including the Start Menu and Settings, became unavailable on the affected computer.

The cause was isolated using the effective AppLocker policy: `Get-AppLockerPolicy -Effective`

The output confirmed that the Packaged app collection was configured for enforcement without an appropriate baseline rule set. Default rules were then created for the Packaged app Rules collection through the AppLocker Group Policy configuration. After the policy was refreshed, the Windows shell operated normally again.

Key Skills Demonstrated

- Active Directory forest and domain design (FSMO roles, replication, RODC deployment).
- Windows Server administration: AD DS, DNS, DHCP failover, Group Policy.
- Security hardening: password policies, auditing, BitLocker recovery, AppLocker enforcement.
- Cross-platform integration: Ubuntu domain join with Samba and Winbind, SMB file-sharing.
- Troubleshooting and diagnostics: dcdiag, repadmin, Get-DhcpServerv4Scope, Get-AppLockerPolicy, wbinfo, smbclient.

10.1. Appendix A - Key Commands and Outputs

Active Directory and Replication

- repadmin /replsummary
- dcdiag
- netdom query fsmo
- Get-ADFineGrainedPasswordPolicy -Filter *
- Get-ADUser <username> -Properties LockedOut

DNS

- nslookup dc01.lab.local

DHCP

- Get-DhcpServerv4Scope -ScopeId 192.168.56.0
- Get-DhcpServerv4ScopeStatistics -ScopeId 192.168.56.0
- Get-DhcpServerv4Failover
- Get-DhcpServerv4Binding
- Set-DhcpServerv4Scope -ScopeId 192.168.56.0 -State Active

Client Networking

- ipconfig /all
- ipconfig /release
- ipconfig /renew

Group Policy

- gpupdate /force
- gpresult /r
- gpresult /r /scope:computer
- gpresult /r /scope:user
- Get-GPO -All
- Get-GPInheritance -Target "OU=_COMPUTERS, DC=LAB, DC=local"
- New-GPLink -Name "AppLocker" -Target "OU=_COMPUTERS, DC=LAB, DC=local"

AppLocker

- Get-AppLockerPolicy -Effective
- \$policy = Get-AppLockerPolicy -Effective
- \$policy.RuleCollections | Select-Object RuleCollectionType, EnforcementMode
- Test-AppLockerPolicy `
 - -Policy (Get-AppLockerPolicy -Effective) `
 - -Path "<file path>" `
 - -User "<username>"

Windows Account Lockout

- Get-ADUser <username> -Properties LockedOut
- Unlock-ADAccount -Identity <username>

Samba and Ubuntu Domain Integration

- net ads join -U Administrator
- net ads testjoin
- wbinfo -t
- wbinfo --online-status
- wbinfo -a <username>
- su - <username>
- smbclient //server/ADShare -U <username>
- testparm -s